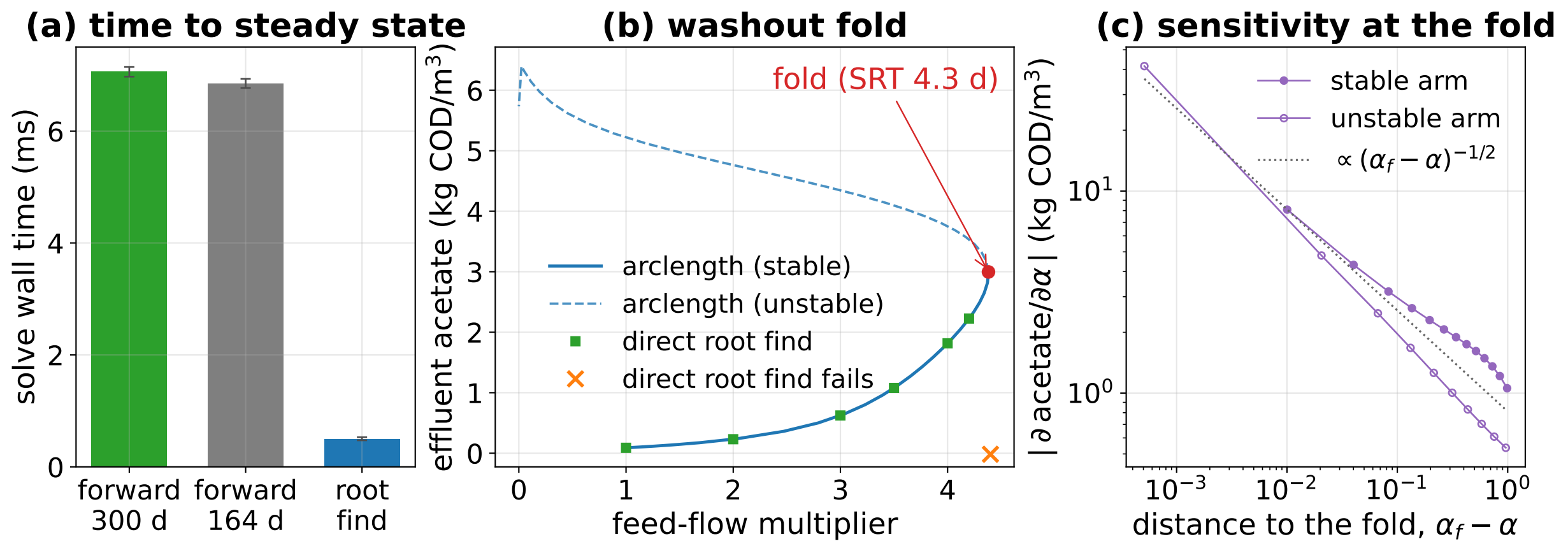
(a) time to steady state
solve wall time (ms)
0
2
4
6
forward 300 d
forward 164 d
root find
(b) washout fold
effluent acetate (kg COD/m^3)
fold (SRT 4.3 d)
arclength (stable)
arclength (unstable)
direct root find
direct root find fails
0
1
2
3
4
5
6
feed-flow multiplier
(c) sensitivity at the fold
$|\partial$ acetate/$\partial\alpha|$ (kg COD/m^3)
stable arm
unstable arm
$\propto (\alpha_f - \alpha)^{-1/2}$
10^1
10^0
10^{-3}
10^{-2}
10^{-1}
10^0
distance to the fold, $\alpha_f - \alpha$